Netflix Content Analysis

MARCH 8

Using Pandas(JUPYTER LAB)

HIMANSHU CHAUHAN

Introduction

Netflix is one of the largest streaming platforms in the world, offering thousands of movies and TV shows across different genres and countries. With the rapid growth of digital streaming, analyzing content trends can provide insights into Netflix's content strategy, global reach, and audience preferences.

This project analyzes the Netflix titles dataset to understand patterns in content distribution, release trends, country contributions, and genre popularity using Python and Pandas.

The objective is to transform raw data into meaningful insights through data cleaning, exploratory data analysis, and visualization.

PROBLEM STATEMENT

The purpose of this project is to analyze Netflix's content catalogue to identify patterns and trends related to:

- **Distribution of movies and TV shows**
- **Growth of content over time**
- **Countries contributing most to Netflix content**
- **Popular genres and categories**
- **Common Audience Ratings**
- **Duration patterns of movies and TV shows**

The goal is to extract insights that help understand Netflix's content ecosystem.

Dataset Description

The dataset used in this project contains information about movies and TV shows available on Netflix.

Dataset Source: Kaggle – Netflix Titles Dataset

The dataset includes several attributes describing each title such as its type, director, cast, country of production, release year, rating, duration, and genre.

Column	Description
show_id	Unique identifier for each title
type	Movie or TV Show
title	Name of the content
director	Director of the content
cast	Actors appearing in the content
country	Country of production
date_added	Date when content was added to Netflix
release_year	Year of original release
rating	Age rating of the content
duration	Length of movie or number of seasons
listed_in	Genre categories
description	Short summary of the content

The dataset contains 8,807 titles and 12 columns describing Netflix movies and TV shows.

Data Cleaning

Real-world datasets often contain inconsistencies and missing values. Several data cleaning steps were performed to prepare the dataset for analysis.

Handling Missing Values

Missing values in columns such as director, cast, and country were handled appropriately to ensure the dataset remained usable for analysis.

Cleaning Country Column

The country column sometimes contained multiple countries in a single entry separated by commas. These entries were separated and cleaned to accurately represent each country involved in content production.

Converting Date Column

The date_added column was converted into datetime format, allowing extraction of additional information such as the year content was added to Netflix.

Handling Duration Column

The duration column contained two types of information:

- **Movie duration in minutes**
- **Number of seasons for TV shows**

The numeric values were extracted from the duration column to allow quantitative analysis.

Exploratory Data Analysis

After cleaning the dataset, several exploratory analyses were performed to understand content patterns on Netflix.

Content Type Distribution

The dataset was analysed to determine the proportion of Movies and TV Shows available on Netflix.

Content Added Per Year

The number of titles added each year was calculated to understand the growth of Netflix's content library.

Country-wise Content Production

Analysis of the country column helped identify which countries produce the most Netflix content.

Genre Distribution

The listed_in column was analysed to determine the most common genres available on the platform.

Duration Analysis

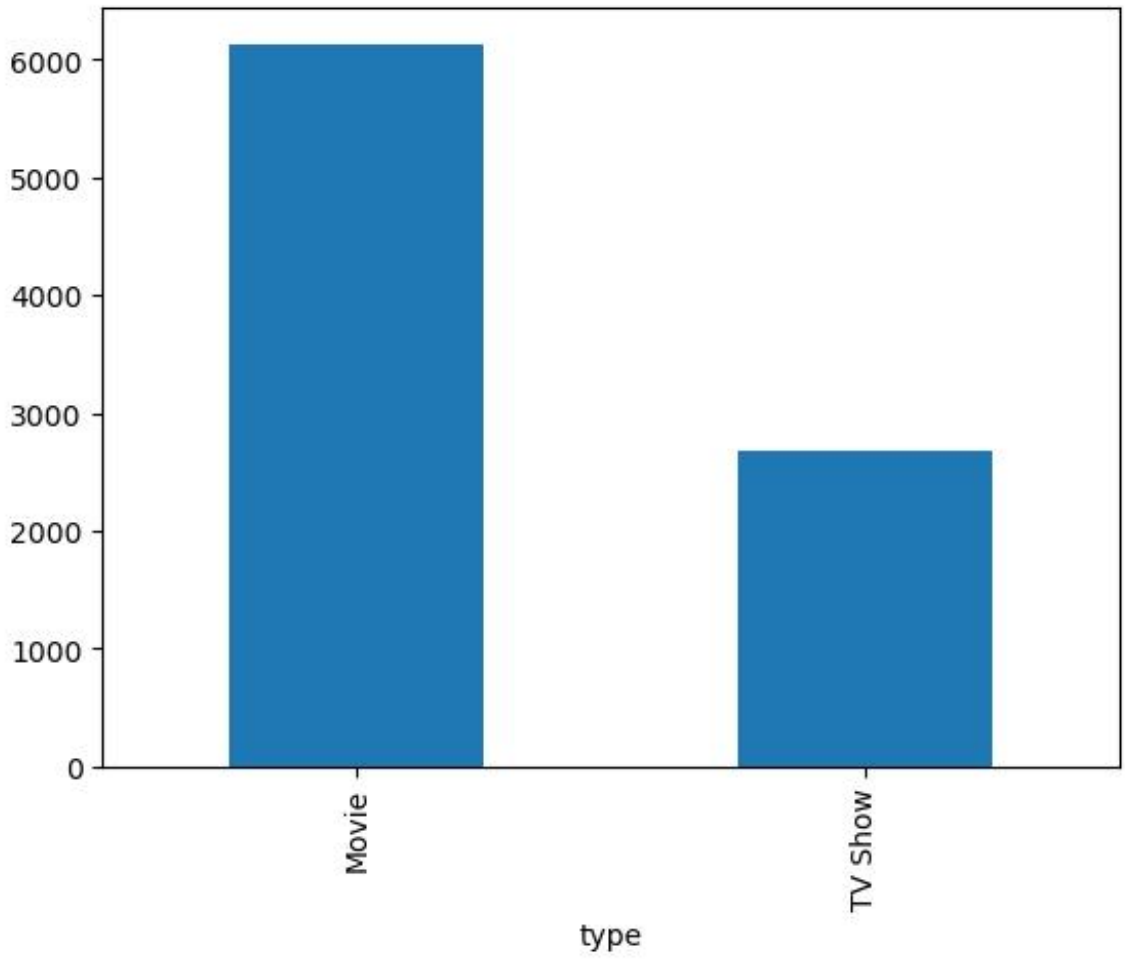
The duration column was used to analyze the distribution of movie lengths and the number of seasons in TV shows.

Key Insights

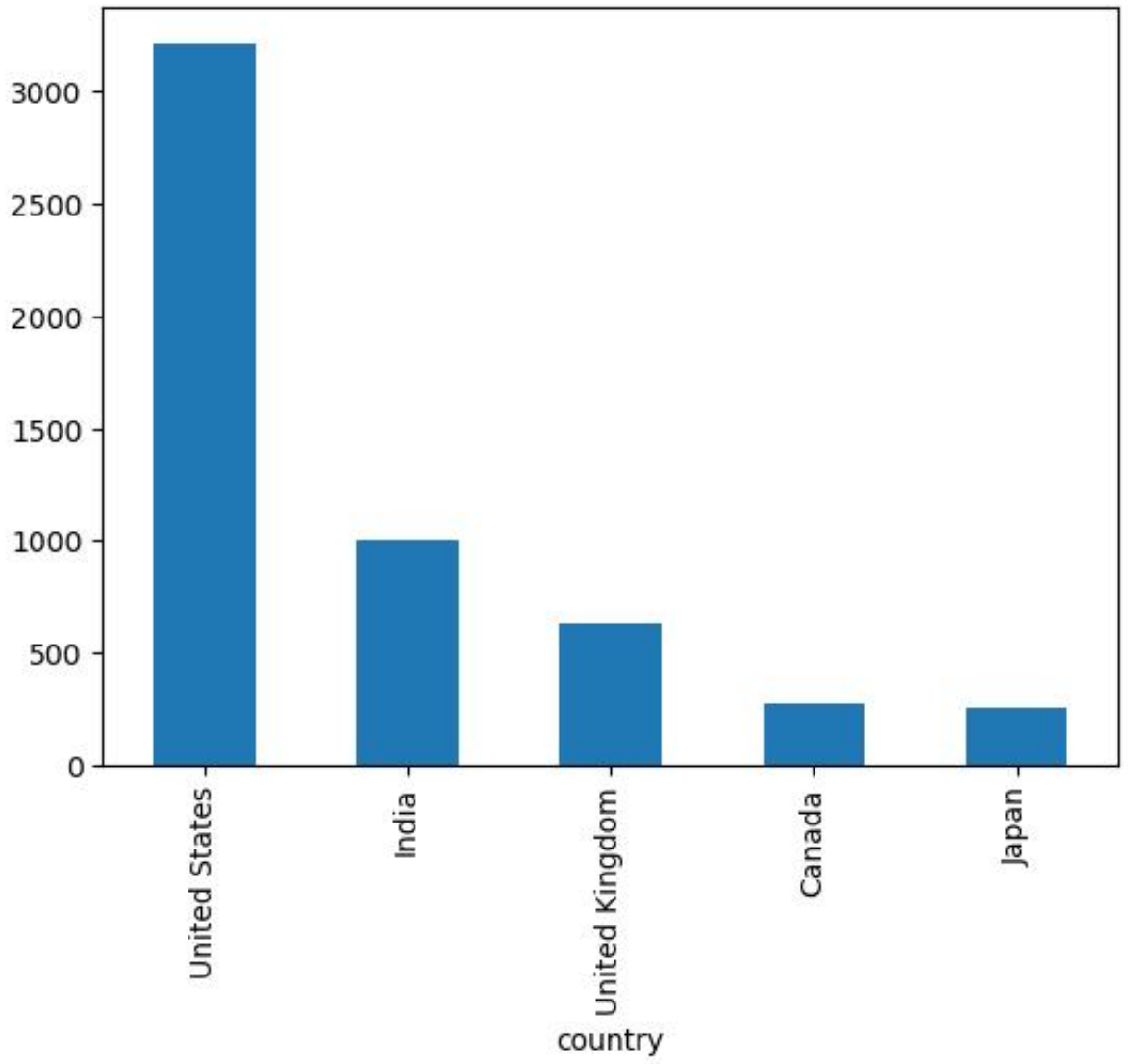
- **The number of Movies is 6131 & that of TV Shows are 2676.**
  The proportion of Movies to TV Shows is around 2.32.
- **United States contribute the most content.**
  The number of titles contributed by United States is 3211.
  Others in the list are, India with 1008 titles and United Kingdom with 628 titles.
- **Netflix experienced significant growth in content additions after 2016.**
- **Drama and Comedies are among the most common genres on the platform.**
- **Most movies on Netflix have durations between 90 and 120.**
- **Most TV shows have one or two seasons.**
- **Most common audience ratings are TV-MA and TV-14.**

Visualizations

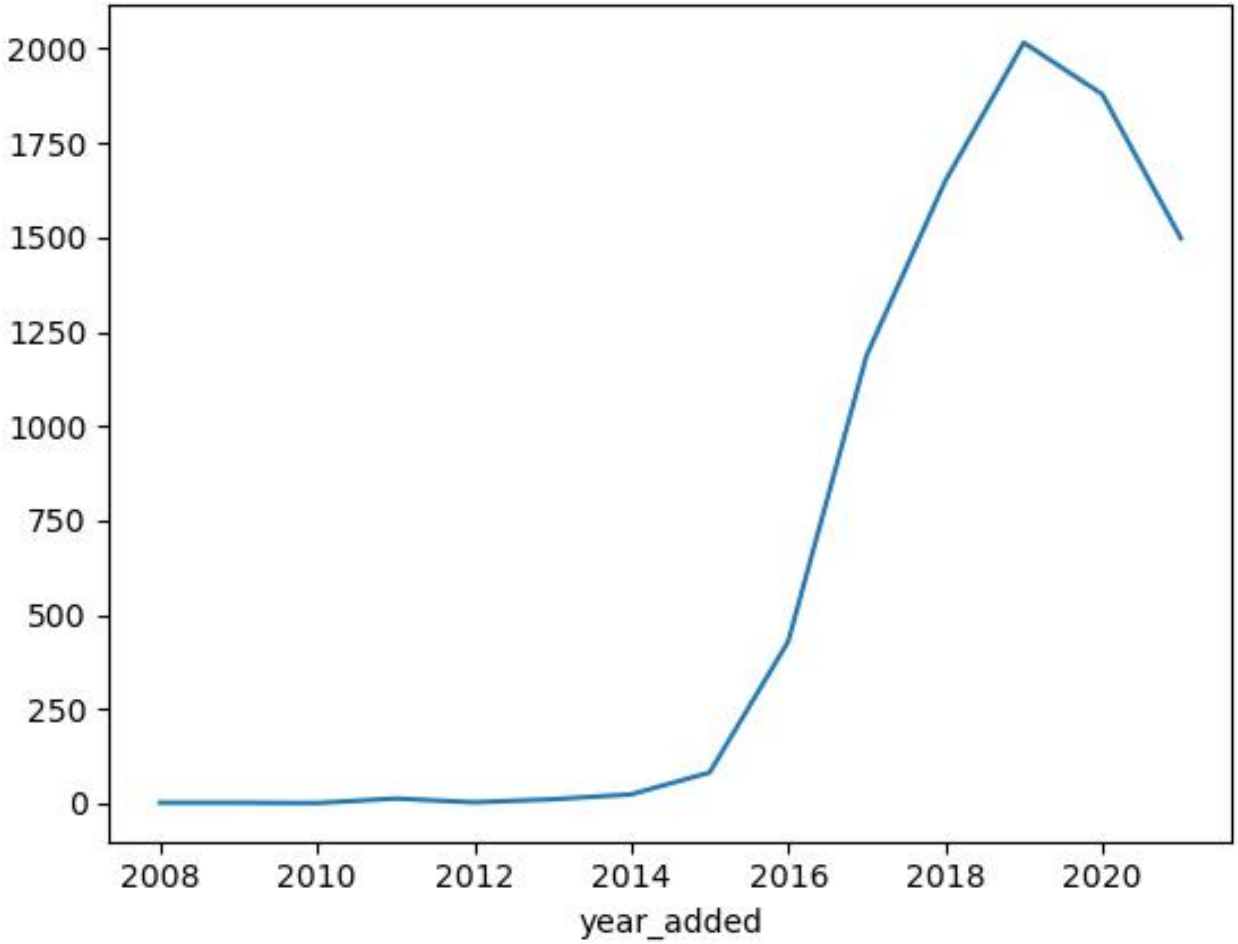
- **Movies vs TV Show Distribution:**



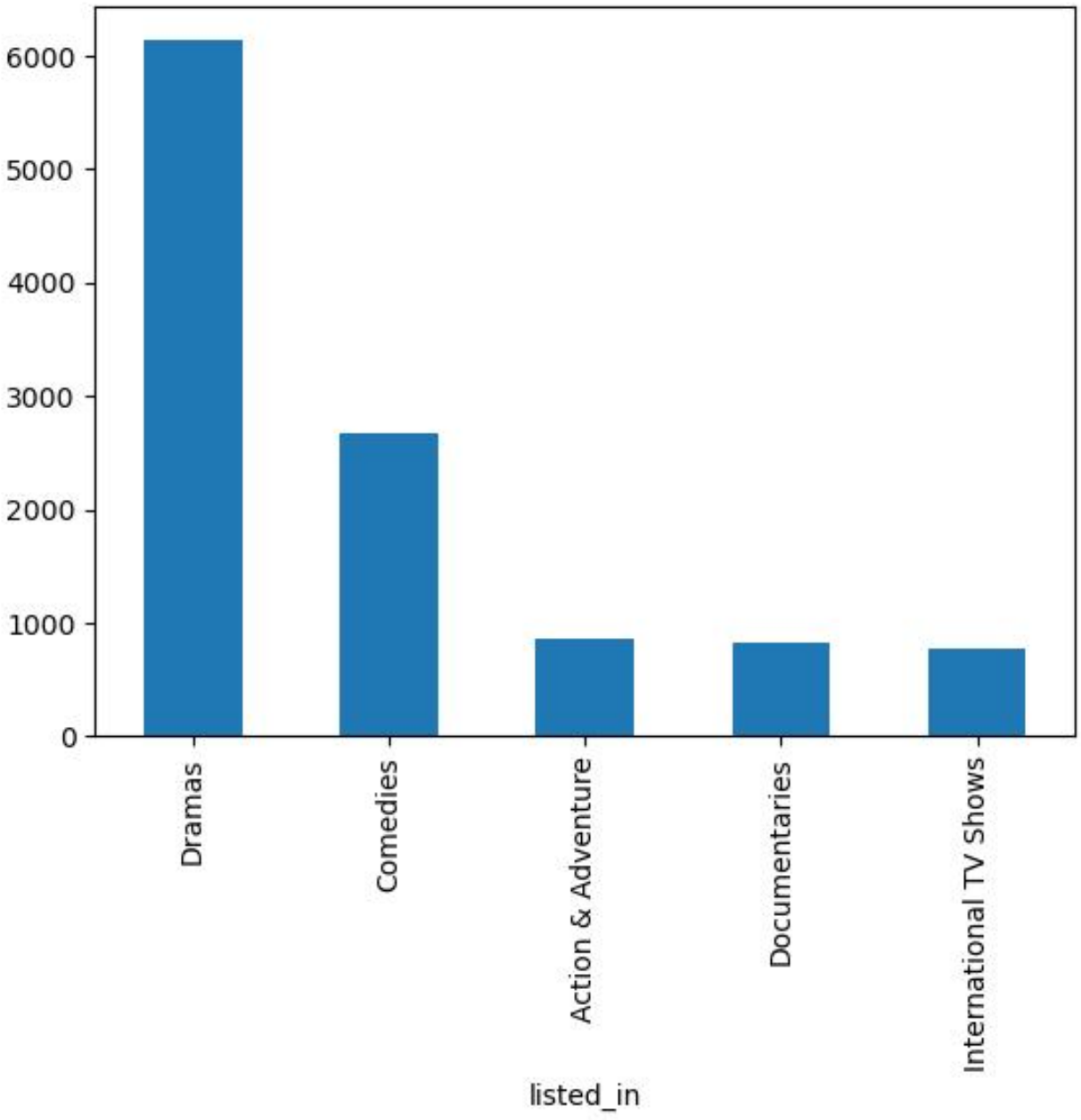
➢ Top producing countries:



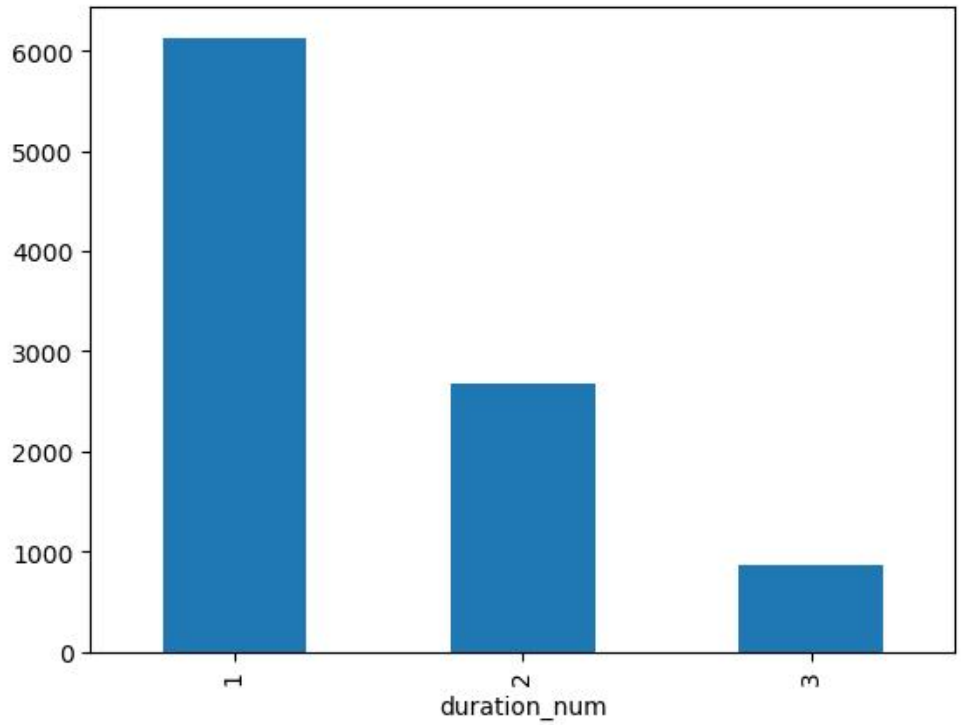
➢ Content added per year (Library Growth):



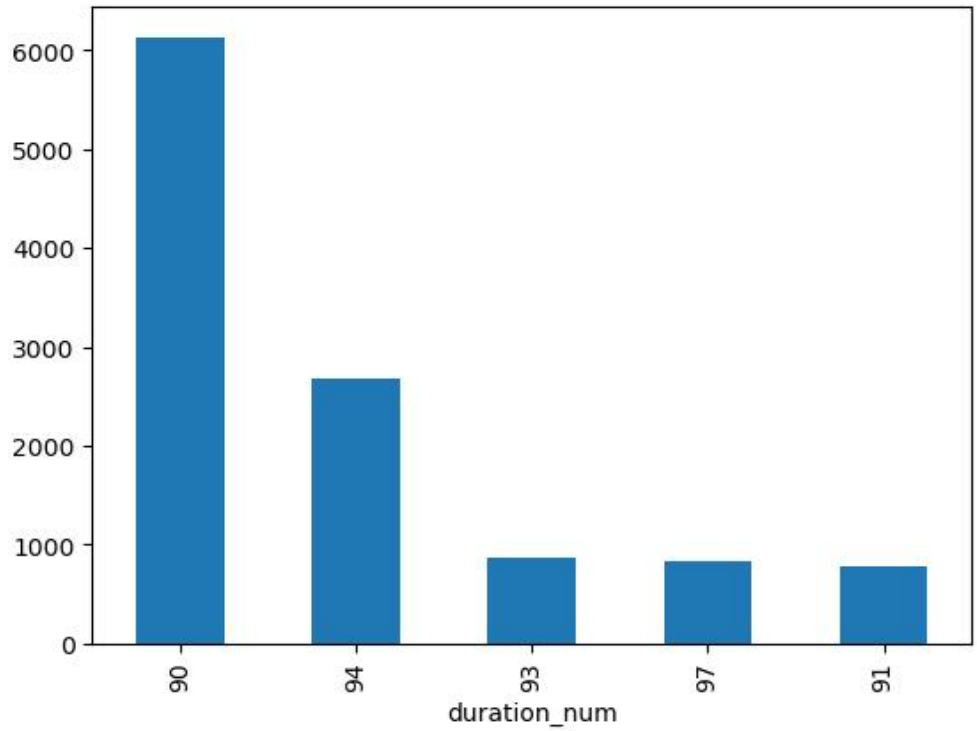
➢ Genre Distribution:



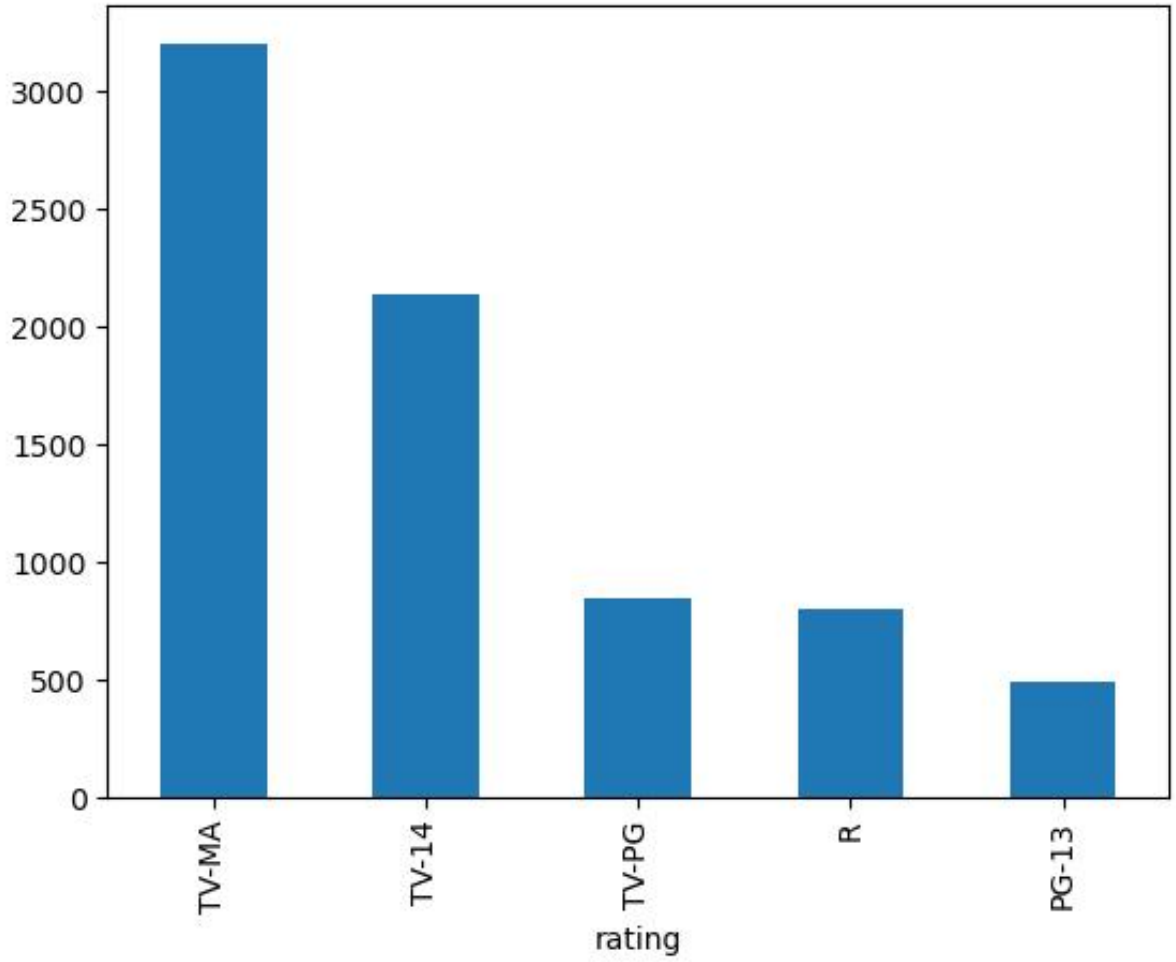
➢ TV Show Duration Distribution:



➢ Movie Duration Distribution:



➢ Common Audience Ratings:



Conclusion

This project explored Netflix's content dataset using Python and Pandas to uncover trends in content production, growth, and distribution.
The analysis revealed key insights into Netflix's global content strategy and highlighted the dominance of movies on the platform. The results also showed how Netflix expanded its content library significantly after 2016 and how different countries contribute to the platform.
Such data analysis techniques can help organizations make informed decisions regarding content production and audience engagement.